

2023



Boosting Performance by 15x Using AWS Elastic Disaster Recovery with Ellucian

Learn how Ellucian built a cohesive disaster recovery service for higher education technology around AWS Elastic Disaster Recovery.

[Overview](#) | [Opportunity](#) | [Solution](#) | [Outcome](#) | [AWS Services Used](#) | [Architecture Diagram](#)

15x

improvement in recovery time objective



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21%

reduction in maintenance costs



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[Ellucian](#) is a leading higher education technology solutions provider, powering the essential operations of colleges and universities worldwide. Its offerings must meet service-level agreements and customer requirements for availability and disaster recovery, which is the restoration of access to vital infrastructure if an unexpected event causes failure.

As it phased out its disaster recovery solution to improve recovery time and reduce operational overhead, Ellucian turned to Amazon Web Services (AWS). Working alongside AWS, Ellucian built an innovative new solution around [AWS Elastic Disaster Recovery](#) (AWS DRS), which minimizes downtime and data loss with fast, reliable recovery of cloud-based applications using affordable storage, minimal compute, and point-in-time recovery. Ellucian's solution uses AWS DRS in combination with AWS serverless solutions and its own application-specific logic. It has improved recovery time objectives (RTO) and recovery point objectives (RPO) by 15 times, reduced maintenance costs by 21 percent, and built a scalable infrastructure to support future growth without a corresponding increase in operational cost.

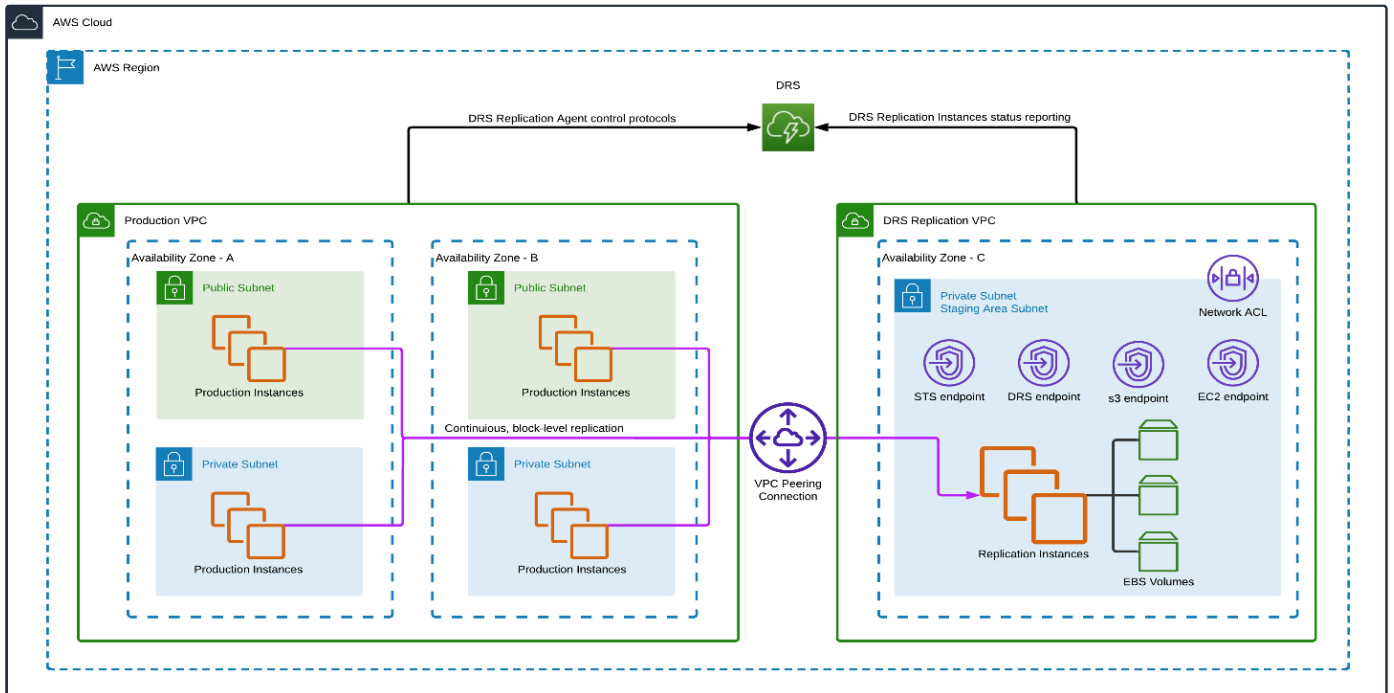
Opportunity | Using AWS Elastic Disaster Recovery for Efficient Disaster Recovery for Ellucian

Founded in 1968, Ellucian supports more than 2,900 higher education institutions across 50 countries, serving 22 million students. Fueled by decades of experience focused on the unique needs of learning institutions, the Ellucian platform features best-in-class software-as-a-service capabilities and the ability to deliver insights for its customers' immediate and future needs. These solutions and services span the entire student lifecycle—including data-rich tools for student recruitment, enrollment, and retention to workforce analytics, fundraising, and alumni engagement.

Increasingly, Ellucian, an [AWS Partner](#) since 2014, has been assisting its customers with digital transformation, migrating legacy business processes and supporting systems to the cloud. "We want our customers to focus on the value-added parts of running an institution," says Blake Keller, Ellucian's director of cloud engineering. "As our customers migrate into the Ellucian cloud built on AWS, they get access to the breadth of things that they can't do for themselves."

Ellucian chose AWS DRS to enhance its disaster recovery solution for its [Amazon Elastic Compute Cloud](#) (Amazon EC2) fleet, a reliable source of secure and resizable compute capacity for virtually any workload. When a failover is initiated, it migrates the affected Amazon EC2 instances and associated block storage to secondary recovery instances in a healthy Availability Zone. "We use AWS DRS to protect the instances and replicate the volumes," says Keller. "But AWS DRS isn't aware of updates or other changes to instances. We wanted to build around that, to gather information about the total configuration of the environment."

Figure 1. Ellucian's disaster recovery reference architecture



Solution | Delivering 15x Improvement in RTO and RPO for Customers

Ellucian combined selected AWS serverless solutions with AWS DRS and added application-specific logic to deliver a cohesive, single-button disaster recovery service for customers. It built a mechanism to regularly gather information about the settings to an instance so that it's exactly as functional as the original instance was. That way, when a failover is initiated, AWS DRS recreates the instance—including updated settings—in the chosen Availability Zone. For example, the solution automatically updates associations with load-balancing target groups, domain name service records, security groups, or internal application connectivity.

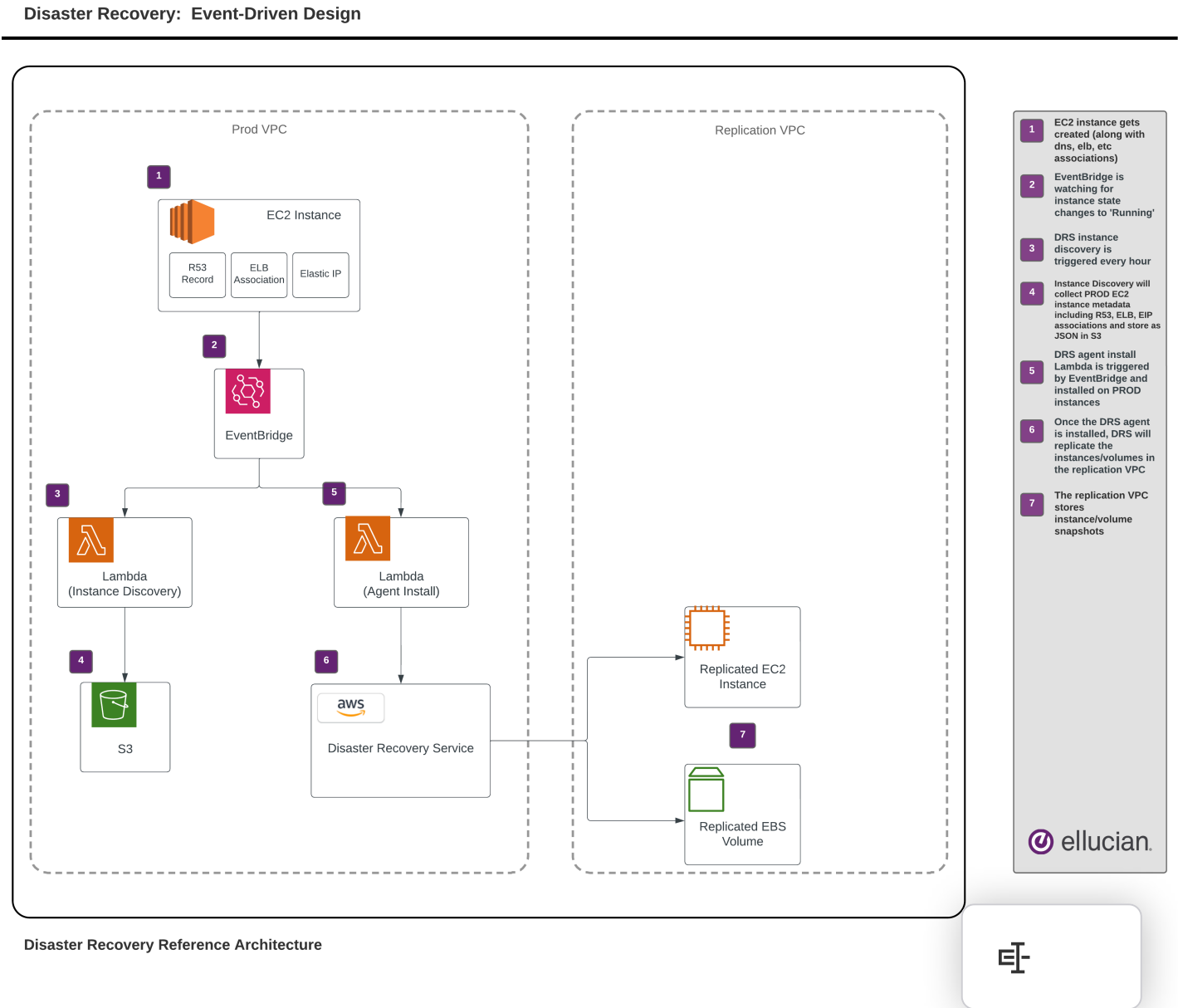
Ellucian is obligated to perform and document yearly disaster recovery tests for customers. In  the company ran its first test for the new service that is based on AWS DRS. The test took under 30 minutes, the fastest

test the company had ever performed and 90 percent faster than its previous solution. "Using AWS DRS, we accomplish the same tests in a fraction of the time, and it doesn't take an army," says Keller. "By extension, we know that we could run an actual disaster recovery failover with the minimum number of resources required."

Engineers or operations personnel have reduced their need to take an active role in providing coverage of instances running within Ellucian's environment. "Using the AWS serverless approach helps us build integrations with a low level of effort, reducing complexity while building scalable, resilient, agile, and cost-effective solutions," says Chris Dooley, lead cloud engineer at Ellucian. Ellucian built an event-driven model using Amazon EventBridge, a serverless event bus that organizations can use to receive, filter, transform, route, and deliver events. Using Amazon EventBridge, the solution automatically connects its instances to AWS DRS using AWS Lambda, a serverless, event-driven compute service that lets organizations run code for virtually any type of application or backend service without provisioning or managing servers. "The use of these services helped us build an event-driven solution that facilitates insight, observability, and maintenance of disaster recovery and dependent services across the entire environment," says Spencer Munjone, lead cloud engineer at Ellucian.

Ellucian has improved both observed RPO and RTO by 15 times each. "No team has to babysit these environments continuously to check on coverage of the disaster recovery service," Keller says. "That is a huge component to making sure that we stay within our RPO and RTO obligations to our customers."

Figure 2. Ellucian's event-driven design



Additionally, Ellucian has reduced its maintenance costs by 21 percent. “Our customers get the peace of mind of knowing that we can do disaster recovery at scale on the world-class cloud solution of AWS,” says Keller. “It’s an overall better experience for them.”

Customers that operate within a single AWS Region have access to multiple Availability Zones, fully isolated partitions of the AWS infrastructure. The solution improved availability to three zones for many of Ellucian’s products. “Using AWS DRS, we have a new level of confidence that we are able to mitigate problems and insulate our customers from outages, thus giving us the ability to commit to our uptime guarantees,” says Keller.

Outcome | Scaling to Accommodate Growth While Minimizing Operational Cost

Ellucian expects to triple its infrastructure over the next few years to accommodate additional customers without a parallel increase in operational cost. “We went into this with one express goal: to have a robust, reliable disaster recovery solution that meets our needs and, by extension, our customers’ needs,” says Keller. “We built a solution that is essentially self-maintaining after deployment, protecting our customers and our resources on AWS and delivering maximum uptime with minimal operational and engineering overhead.”

About Ellucian

As a marketplace leader in higher education technology, Ellucian works with more than 2,900 customers in 50 countries, serving 22 million students. With deep experience and a singular focus on learning institutions, the Ellucian software-as-a-service platform delivers insights needed now and in the future.

AWS Services Used

AWS DRS

AWS Elastic Disaster Recovery (AWS DRS) minimizes downtime and data loss with fast, reliable recovery of on-premises and cloud-based applications using affordable storage, minimal compute, and point-in-time recovery.

[Learn more »](#)

Amazon EC2

Amazon EC2 provides secure, resizable compute in the cloud, offering the broadest choice of storage, networking, OS, and purchase model.



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Amazon EventBridge

Amazon EventBridge is a serverless event bus that ingests data from your own apps, SaaS apps, and AWS services and routes that data to targets.

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AWS Lambda

AWS Lambda is a serverless, event-driven compute service that lets you run code for virtually any type of application or backend service without provisioning or managing servers.

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The University of Applied Sciences and Arts Northwestern (FHNW) is one of Switzerland's largest universities. In collaboration with interactive museum for analog and digital technology Enter Technikwelt Solothurn, FHNW embarked on a generative artificial intelligence (AI) project using Amazon Web Services (AWS). With managed service Amazon SageMaker, says research associate Claudio Paonessa,

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